

Lecture 6

*Fourier Analysis:
from spatial to frequency domain*

fjv, image analysis in microscopy 2006

1

Content Lecture 6

- What have we seen from Fourier so far
- Fourier transform
 - Continuous, Discrete
- Magnitude and Phase
- Power Spectrum
- Padding
- Filters
 - Low pass
 - High pass
 - Band pass
- Optics
- Homomorphic Filter, Notch Filter

fjv, image analysis in microscopy 2006

2

Joseph Fourier

- Baron
Jean Baptiste Joseph Fourier
- 1768-1830, France
- Mathematician



fjv, image analysis in microscopy 2006

3

Initial Application Fourier Analysis

- “Theorie analytique de la chaleur”
- 1822: study in heat propagation:
decomposition of propagation in a
number of sine and cosine terms
- Now known as “Fourier Analysis”
and applied in many fields

fjv, image analysis in microscopy 2006

4

Applications in Image Processing

To name a few ...

- Image filtering
- Image enhancement, Noise Suppression
- Image restoration
- Image compression
- Image registration
- Texture analysis
- Description of Optics

fjv, image analysis in microscopy 2006

5

Paradigms in Image Processing

- Signal processing (linear system theory)
 - A signal, normally 1 dimensional
 - Signal is sampled in 2 orthogonal dimensions: image
 - ...or in more orthogonal dimensions: (3D, 4D)
 - Fourier Transform is quintessential in understanding the image signal (decomposition)
- Set theory.
 - Concepts of mathematical morphology
 - Uses structural elements to process an image.

fjv, image analysis in microscopy 2006

6

Fourier Transform

- The FT is based on a set of generalized harmonics.
- These harmonics form an ortho-normal basis set for functions:

$$\{e^{i2\pi st}\}$$

where each harmonic has a different frequency s

- Euler relation: $e^{i2\pi st} = \cos(2\pi st) + i \sin(2\pi st)$
- In the Fourier transform we decompose an image into **sine** and **cosine** components. Each point in the FT represents a frequency

fjv, image analysis in microscopy 2006

7

2D Continuous Fourier Transform

Basic functions are sinusoids with frequency u in one direction times sinusoids with frequency v in the other:

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-i2\pi(ux+vy)} dx dy$$

Same process for the inverse transform:

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{i2\pi(ux+vy)} dx dy$$

fjv, image analysis in microscopy 2006

8

Facts

- Images have a finite size
- This finite size is described by MxN matrix
- MxN matrix is referred to as the image size
- Image consist of samples
- In stead of the continuous Fourier Transform the Discrete Fourier Transform (DFT) is used.
- DFT is the Fourier Transform on a sampled function

fjv, image analysis in microscopy 2006

9

2D Discrete Fourier Transform

For an NxN image, the Discrete Fourier transform is:

$$F[u, v] = \frac{1}{N^2} \sum_{x,y=0}^{N-1} f[x, y] \cdot e^{-j2\pi(xu+yv)/N}$$

For an NxM image, the Discrete Fourier transform is:

$$F[u, v] = \frac{1}{NM} \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} f[x, y] \cdot e^{-j2\pi(xu/N+yv/M)}$$

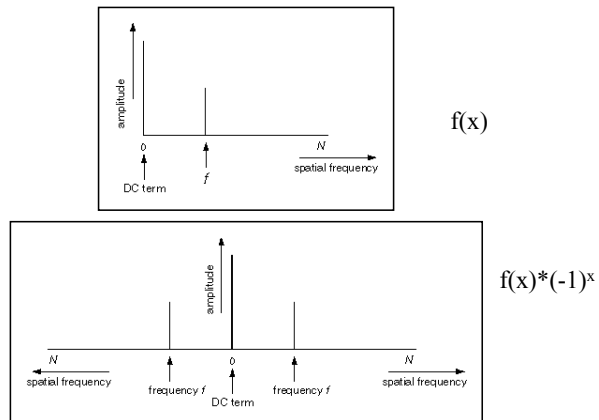
For an NxN image, the reverse Discrete Fourier transform is:

$$f[x, y] = \frac{1}{N^2} \sum_{u,v=0}^{N-1} F[u, v] \cdot e^{j2\pi(xu+yv)/N}$$

fjv, image analysis in microscopy 2006

10

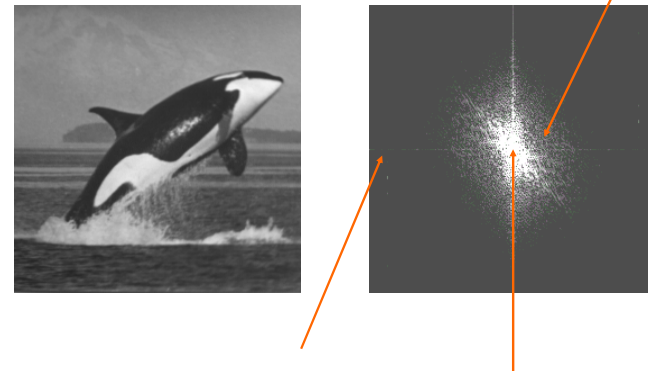
1D Fourier Transform



fjv, image analysis in microscopy 2006

11

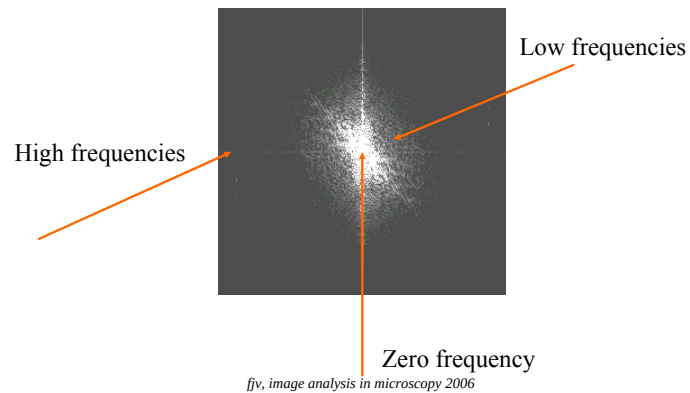
Applying the Fourier Transform



fjv, image analysis in microscopy 2006

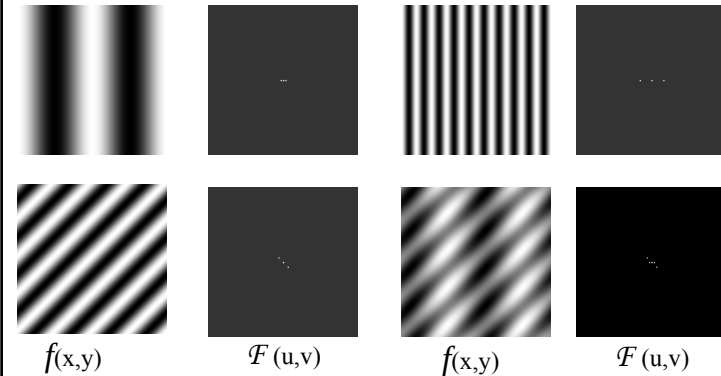
12

Analyzing the Fourier Transform



13

Straightforward examples



Images from Dr. S. Lehar

fjv, image analysis in microscopy 2006

14

Components of Fourier Transform

- Image with float numbers
 - One representing the real part
 - One representing the imaginary part
- Zero frequency: **DC component**, at $F(0,0)$
 - DC: direct current
 - Represents average brightness value in the image
- Low frequencies: slowly varying components
- High frequencies: fast varying components
- Normally the modulus of the amplitude shown for display

fjv, image analysis in microscopy 2006

15

Representation complex numbers

- Complex numbers can be dealt with as:
 - *real, imaginary*
- Or
 - *magnitude, phase*
- Interpretation:
 - Real:** How much of a cosine of certain frequency you need
 - Imaginary:** How much of a sine of certain frequency you need
 - Magnitude:** Amplitude of combined cosine and sine
 - Phase:** Relative proportions of sine and cosine

fjv, image analysis in microscopy 2006

16

Magnitude & Phase

Magnitude is defined as: $|F| = \sqrt{\Re(F)^2 + \Im(F)^2}$

Phase is defined as: $\phi(F) = \arctan\left(\frac{\Re(F)}{\Im(F)}\right)$

Real: How much of a cosine of certain frequency you need

Imaginary: How much of a sine of certain frequency you need

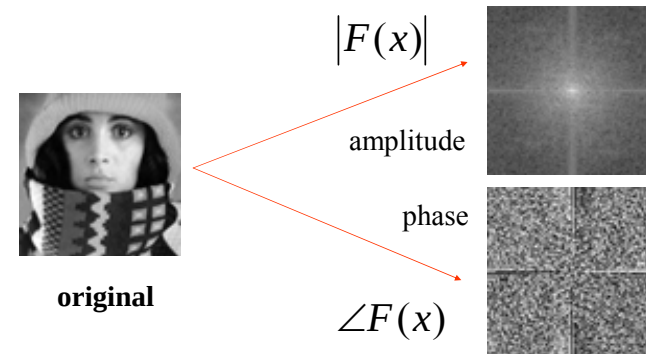
Magnitude: Amplitude of combined cosine and sine

Phase: Relative proportions of sine and cosine

fjv, image analysis in microscopy 2006

17

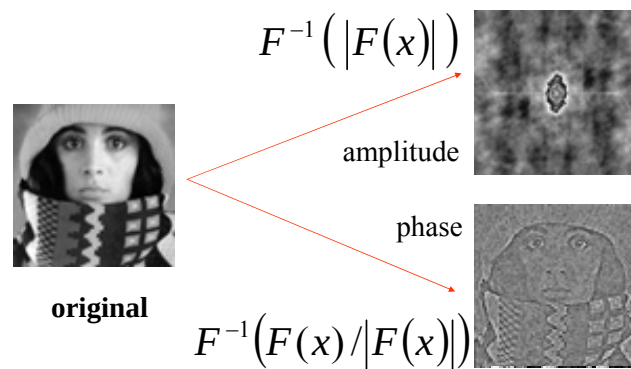
Phase and Magnitude (amplitude)



fjv, image analysis in microscopy 2006

18

Phase and Magnitude (amplitude)



fjv, image analysis in microscopy 2006

19

Power Spectrum

- Multiply with complex conjugate
- In fact, the power spectrum of a signal is the Fourier Transform of its autocorrelation function:

$$\begin{aligned} P(s) &= \mathcal{F}(f(t) * f(-t)) \\ &= F(s) F^*(s) \\ &= |F(s)|^2 \end{aligned}$$

- This equals (also) the squared magnitude of the Fourier transform of the function
- It is entirely real (no imaginary part)
- Due to complex conjugate multiplication

fjv, image analysis in microscopy 2006

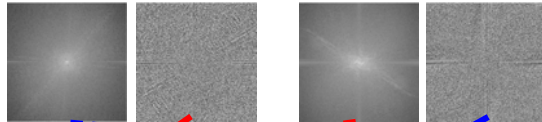
20

Information in Phase Component

Original Images

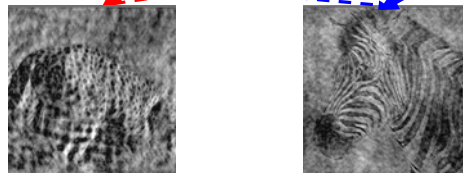


Magnitude (L)
and
Phase (R)



Reconstruct
by FFT^{-1} :
mixing the
magnitude and
phase contribution

Phase "Wins"



fjv, image analysis in microscopy 2006

21

Fast Fourier Transform

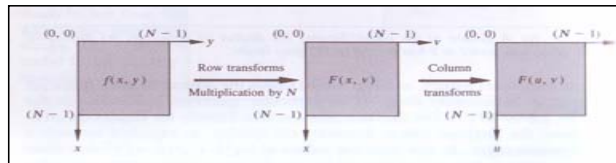
- The computational complexity of the DFT is $O(N^2)$
- Fast Fourier Transformation uses the fact that 2D FT can be separated in 1D cases.
- Also known as Cooley-Tukey algorithm
- First the rows and then the columns
- The computational complexity of the FFT is $O(N \log N)$
- Starting point is a 2^n sized image (e.g. 256×256)
- The result of FFT does not differ from DFT
- All examples are based on FFT

fjv, image analysis in microscopy 2006

22

Separable DFT: FFT

- First the rows are transformed
 - $N/2$, then $N/4$, $N/8$ etc.
- Second the columns are transformed
 - $N/2$, then $N/4$, $N/8$ etc.
- Third results are added.
- Cooley-Tukey algorithm: divide & conquer strategy



fjv, image analysis in microscopy 2006

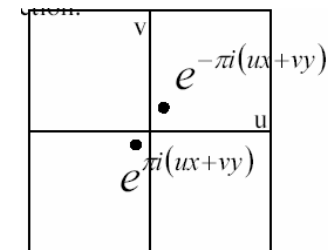
23

Periodicity



$f(x, y)$

$f(x, y) * (-1)^{x+y} \longrightarrow F(u, v)$



fjv, image analysis in microscopy 2006

24

Analysis of FFT 1-D data.

Size Data Set	DFT Multiplications	DFT Time	FFT Multiplications	FFT Time
1024	1 E 6	1 sec	10,240	0.01 sec
8192	67 E 6	67 sec	106,496	0.1 sec
65536	4 E 9	71 min	1,048,576	1.0 sec
1048576	1 E 12	305 hr	20,971,520	20.9 sec

1 complex multiplication takes 1 μ s

Source: National Instruments, Developer Zone

fjv, image analysis in microscopy 2006

25

Analysis of FFT 2-D data.

Image Size	DFT Multiplications	DFT Time	FFT Multiplications	FFT Time
256 X 256	4.3 E 91	71 min	1,048,576	1.0 sec
512 X 512	6.8 E 10	19 hr	4,718,592	4.8 sec
1024 X 1024	1.1 E 12	12 days	20,971,520	21.0 sec
2048 X 2048	1.8 E 13	203 days	92,274,688	92.2 sec

1 complex multiplication takes 1 μ s

Source: National Instruments, Developer Zone

fjv, image analysis in microscopy 2006

26

Padding, Zero-Padding

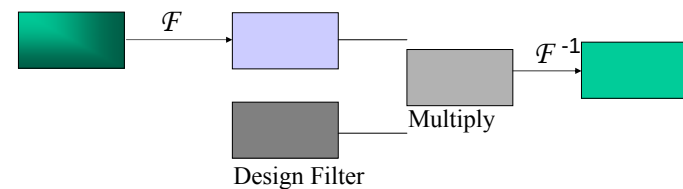
- What to do near the image borders.
- Image considered as a self replication at the borders (torus)
- Adding average value.
- Adding zero-value (zero-padding).
- Padding that is chosen depends on application.
- Padding allows the FFT to non 2^n sized images
- FFT treats a signal (image) as periodic.

fjv, image analysis in microscopy 2006

27

Filtering in Frequency Domain

- Recipe:

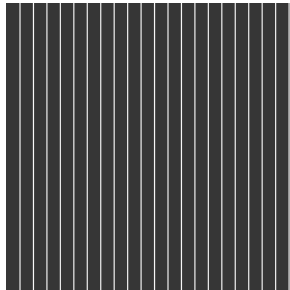


- This is applied on 2^N sized example images
– 256x256

fjv, image analysis in microscopy 2006

28

1st step: forward transform



$f(x,y)$

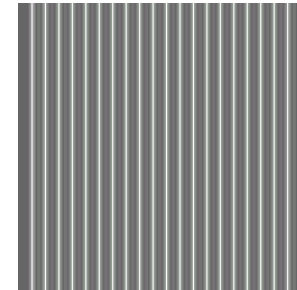
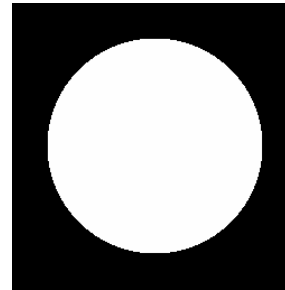


$F(u,v)$

fjv, image analysis in microscopy 2006

29

2nd step: applying low-pass filter

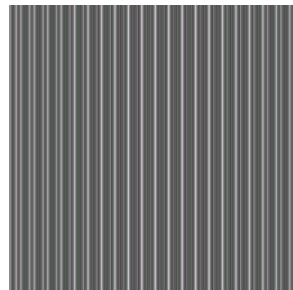
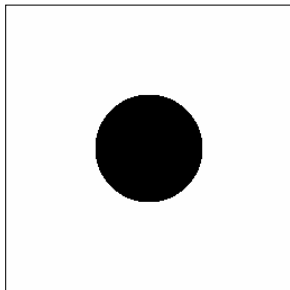


$f'(x,y) = F^{-1}(u,v)$

fjv, image analysis in microscopy 2006

30

2nd step: applying high-pass filter



$f'(x,y) = F^{-1}(u,v)$

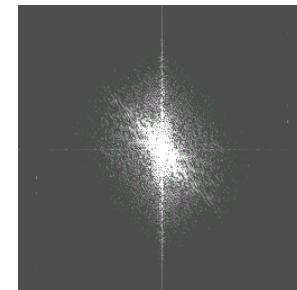
fjv, image analysis in microscopy 2006

31

Filtering in frequency domain



$f(x,y)$

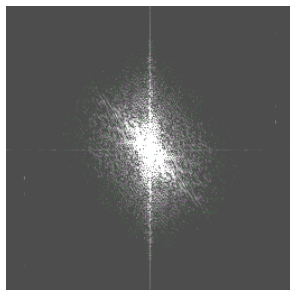


$F(u,v)$

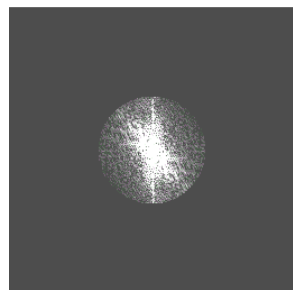
fjv, image analysis in microscopy 2006

32

Low-pass filter, sharp cut-off



$F(u,v)$



$F'(u,v)$

fjv, image analysis in microscopy 2006

33

Reverse Transform



$f(x,y)$

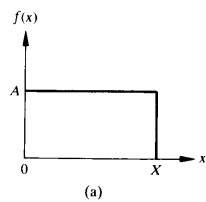


$f'(x,y)$

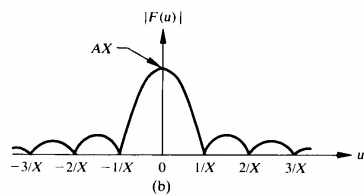
fjv, image analysis in microscopy 2006

34

Ringing



(a)



(b)

A rectangle transforms as a sinc-function: $\text{sinc}(a\pi s) = \frac{\sin(a\pi s)}{a\pi s}$

fjv, image analysis in microscopy 2006

35

Low-pass filter, smooth cut-off



$f(x,y)$



$f'(x,y)$

fjv, image analysis in microscopy 2006

36

High-pass filter, sharp cut-off



$f(x,y)$

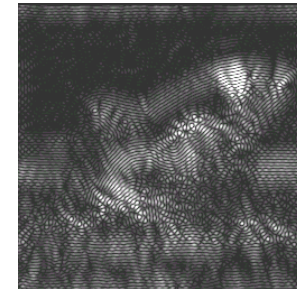
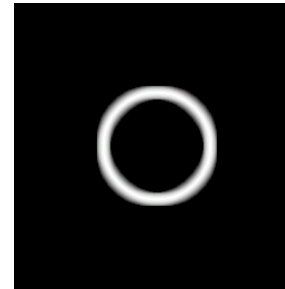


$f'(x,y)$

fjv, image analysis in microscopy 2006

37

Band-pass filter, smooth cut-off



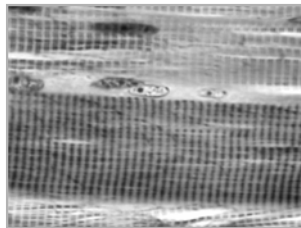
$f'(x,y)$

fjv, image analysis in microscopy 2006

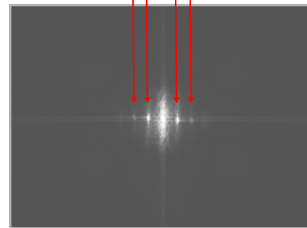
38

Periodic Patterns in FFT

Striated Skeletal Muscle



$f(x,y)$



$F(u,v)$

fjv, image analysis in microscopy 2006

39

Auto-Correlation

- Autocorrelation is the correlation of a function with itself:
- $$f(t) * f(-t)$$
- This is also known as the Power Spectrum

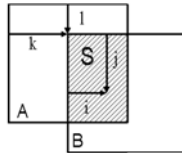
fjv, image analysis in microscopy 2006

40

Spatial Correlation

- Position of the maximum determined by a convolution kernel to find a shift between two images.

$$\rho(k,l) = \frac{\sum_{i,j \in S} A_{(i+k,j+l)} B_{(i,j)}}{\sqrt{\left[\sum_{i,j \in S} (A_{(i+k,j+l)})^2 \right]} \sqrt{\left[\sum_{i,j \in S} (B_{(i,j)})^2 \right]}}$$



fjv, image analysis in microscopy 2006

41

Correlation

Convolution is $f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$

Correlation is $f(t) * g(-t) = \int_{-\infty}^{\infty} f(\tau)g(t+\tau)d\tau$

fjv, image analysis in microscopy 2006

42

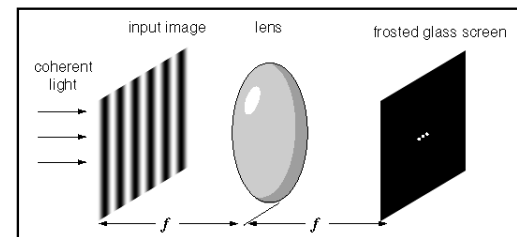
Correlation in Frequency domain

- Auto-correlation.
- A shift of an image results in a shift of the maximum of the auto-correlation.
- This shift corresponds to the translation vector applied to the image.
- Convolution is simplified to 1 conjugate multiplication
- Transforming the image to a ϕ, r image can resolve rotation in the same manner; it will find ϕ .
- Transformation is referred to as polar re-sampling.

fjv, image analysis in microscopy 2006

43

Optics and Fourier (1)

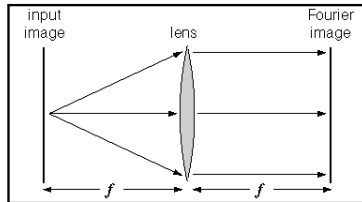


- experiment to show that lens applies a Fourier Transform
- grating in front of lens
- frosted glass in focal plane: Fourier transform of grating (image)

fjv, image analysis in microscopy 2006

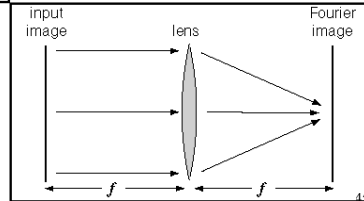
44

Optics and Fourier (2)



- Parallel beam.
- Point input image is spread uniformly over the Fourier image
- Constructive and destructive interference will automatically produce the proper Fourier representation.

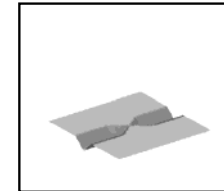
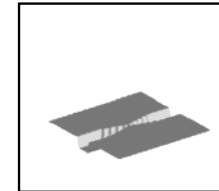
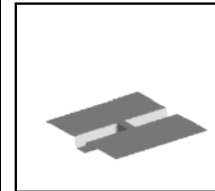
- Parallel rays are focused onto the single central point of the Fourier image,
- Defines the central DC term as the average brightness of the input image.



fjv, image analysis in microscopy 2006

45

Notch filters



- Filter out a certain range of “unwanted” frequencies
- Not affecting the contrast in the image
- Prevent introducing high frequencies: ringing
- Designed for application

fjv, image analysis in microscopy 2006

46

Application of the Notch Filter

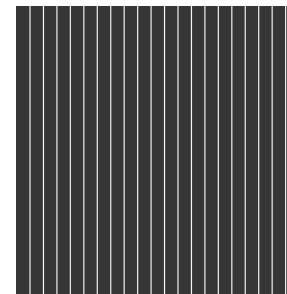


- Application allows easy and straightforward segmentation
- Here notch filter is de facto a restoration; some interference occurred at the acquisition phase.

fjv, image analysis in microscopy 2006

47

Infinite higher order terms



$f(x,y)$



$F(u,v)$

fjv, image analysis in microscopy 2006

48

Homomorphic Filter

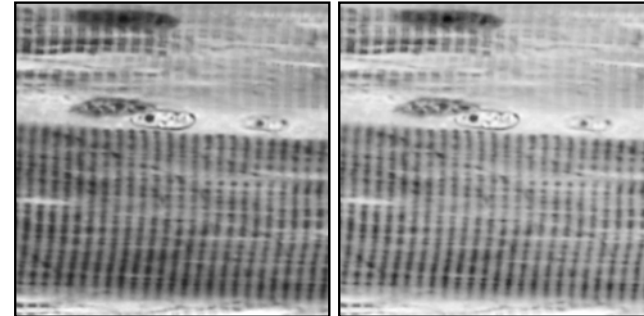
- Separation of illumination and reflectance components: $f(x,y) = i(x,y) \cdot r(x,y)$
- Normally, Illumination and Reflectance are not separable.
- $z(x,y) = \ln(f(x,y))$
 $\quad = \ln(i(x,y)) + \ln(r(x,y))$
- In like manner we see:
 $\mathcal{F}\{z(x,y)\} = \mathcal{F}\{\ln(f(x,y))\}$
 $\quad = \mathcal{F}\{\ln(i(x,y))\} + \mathcal{F}\{\ln(r(x,y))\}$

fjv, image analysis in microscopy 2006

49

Application Homomorphic Filter

Filter with default settings



$f(x,y)$

$f'(x,y)$

fjv, image analysis in microscopy 2006

50

Other transforms

- | | |
|---------------------|--------------------------------|
| • Cosine transform | • Cosine functions |
| • Haar transform | • Hierarchical block functions |
| • Wavelet transform | • Wavelets |
| • Hartley transform | • Homomorphic filtering |

fjv, image analysis in microscopy 2006

51

Review Lecture 5



- Continuous FT
- Discrete FT
- Components of FT image
- Fast FT mostly used in Image Processing
- Filtering in frequency domain
- Autocorrelation and Correlation
- Restoration

fjv, image analysis in microscopy 2006

52

Acknowledgements

Books

- Gonzales & Woods: Digital Image Processing
- Holst, Sampling: Aliasing and Data Fidelity

Materials

- Lehar: notes
- FJV, test and experimental data